01 • TITLE SLIDE & EXECUTIVE OVERVIEW

TRL 7 • Working Product & Pilot-Ready Platform

ResQ-Gov

Resilient Emergency Resource Allocation & Evacuation Logistics Governance Platform

"Right Aid, Right Time, Zero Leakage: Enforcing Public-Sector Disaster Accountability via Code-Level Segregation of Duties, Demographic Quota Mathematics, Dual-Key 6-Digit Handover OTP, and Cryptographic Hash-Chain Audit Trails."

COMPETITION CATEGORY	PRODUCT MATURITY	ACCEPTED SYSTEM FORMAT	TARGET ENTITIES
02 Public Sector & Gov	**Working Product (MVP+)**	**Web & Cloud Platform**	**Dinas Sosial & BPBD**
Digital Gov • Citizen Services	Ready for Pilot Deployment	Enterprise GovTech System	514 Regencies/Cities & BNPB

Executive Summary

PUBLIC SECTOR ICT

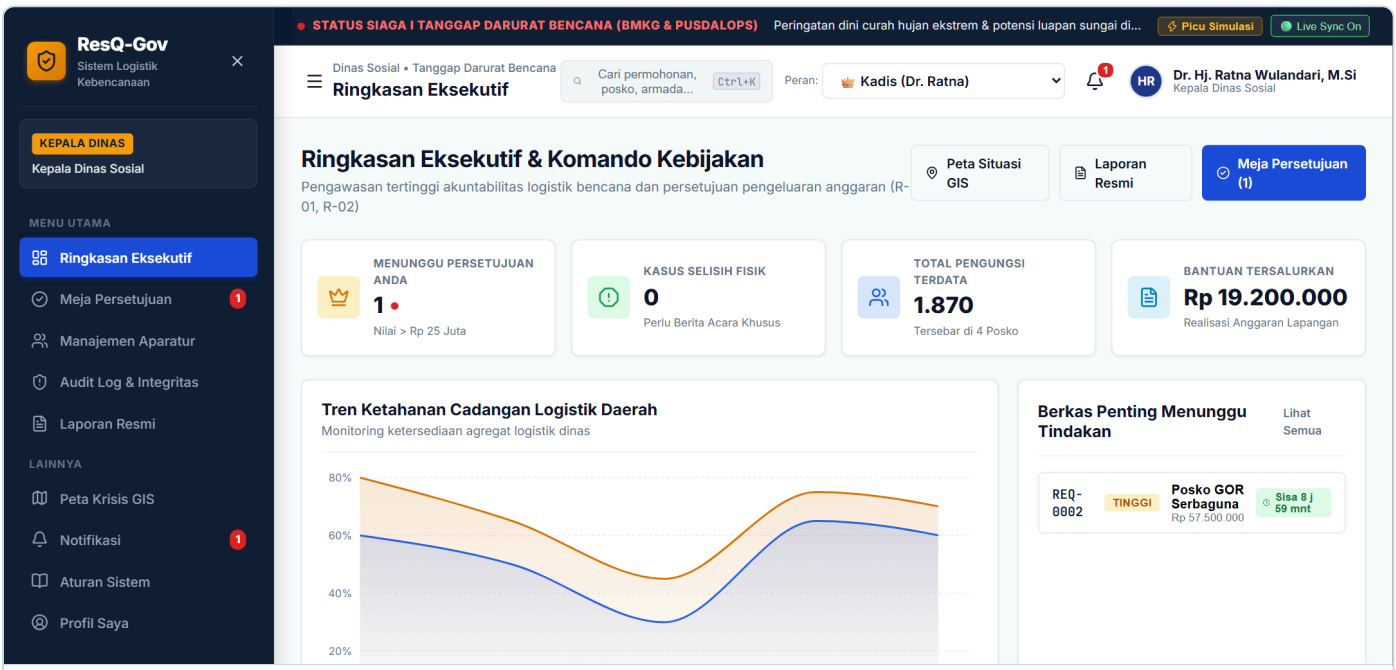
ResQ-Gov is a mission-critical Digital Government command and logistics platform engineered for Regional Social Services Agencies (*Dinas Sosial*) and Disaster Management Agencies (*BPBD*). Unlike conventional government portals that function merely as passive data-entry spreadsheets, ResQ-Gov embeds **12 Binding Public-Sector Governance Rules (R-01 to R-12)** directly into its state-transition engine.

By combining **Segregation of Duties (SoD)**, **Per-Capita Demographic Quota Capping**, **Zero-Trust 6-Digit Handover OTPs**, and an **Append-Only Cryptographic Hash-Chain Audit Trail**, ResQ-Gov eliminates phantom aid requisitions, prevents political hoarding, accelerates emergency dispatch within a 1-to-6-hour SLA, and guarantees 100% auditability for Supreme Audit Institution (*BPK RI*) inspections.

Four-Pillar Governance Architecture

APICTA STANDARD

- **Pillar 1 — Executive Command & Multi-Tier Approval (R-01, R-02):** Enforces strict actor separation between Camp Requestor, Warehouse Verifier, and Head of Agency Approver (mandatory for requisitions > IDR 25,000,000).
- **Pillar 2 — Mathematical Quota & Stock Reservation (R-03, R-04):** Caps aid requests dynamically based on verified camp headcounts (max 3-day consumption norm) minus in-flight quantities.
- **Pillar 3 — Last-Mile Custody & Discrepancy Detection (R-05, R-06):** Requires a secret 6-digit camp OTP upon truck arrival (3-strike lock) followed by itemized physical count verification.
- **Pillar 4 — Tamper-Evident Forensic Ledger (R-09, R-10):** Chains every state mutation into an immutable hash sequence (`prevHash → hash`), instantly exposing any database tampering.



Live Working Product Preview — ResQ-Gov Executive Command Cockpit (Kepala Dinas / Head of Agency View)
Real-time monitoring of regional stockpile resilience, high-value approval queues (> IDR 25M), physical discrepancy alerts, active evacuee census (1,870 citizens across 4 camps), and SIAGA I hydrometeorological early warnings from BMKG & Pusdalops.

12 Rules	≤ 1 Hr / 6 Hr	0% Leakage	100% Audit
Code-Enforced Governance	**Emergency Verify & Deliver SLA**	**Dual-Key 6-Digit Handover OTP**	**Cryptographic Hash-Chain**
R-01 to R-12 hard validation	Down from 48–72 hrs manual cycle	3-strike lockdown + BAST discrepancy	Tamper-evident BPK RI compliance

02 Problem Statement

Why Legacy Disaster Relief Administration Fails Citizens During the Critical 72-Hour Golden Window

2.1 Empirical Context: The Indonesian Disaster Logistics Paradox

Situated along the Pacific Ring of Fire and subject to intensifying monsoon extremes, Indonesia records over **3,000 natural disasters annually** (BNPB Disaster Data, 2024–2026), displacing millions of citizens across floods, landslides, earthquakes, and volcanic eruptions. During the first 72 hours (the humanitarian *Golden Window*), survival depends not merely on the existence of national relief stockpiles, but on the **speed, mathematical equity, and custodial integrity of last-mile distribution** executed by Regional Social Services Agencies (*Dinas Sosial*) and Regional Disaster Management Agencies (*BPBD*).

Paradoxically, post-disaster evaluations consistently reveal that while district warehouses (*Gudang Induk*) hold sufficient buffer stocks of rice, medicines, and shelter kits, field evacuation camps (*Posko Pengungsian*) suffer simultaneous **acute stockouts, severe over-hoarding, and unverified cargo disappearance**. Existing government software tools fail because they are built as passive reporting forms rather than **rule-enforcing governance engines**.

2.2 Four Root Causes of Public Sector Logistics Failure

1. Information Asymmetry & Sensus Blindspots — LATENCY

Field officers report evacuee headcounts via fragmented WhatsApp messages and paper tables. Command centers lack real-time visibility into camp saturation or vulnerable sub-populations (infants needing MP-ASI formula, elderly requiring chronic medication, pregnant women, and persons with disabilities). Stale census data leads to severe supply mismatches.

2. Panic Hoarding & "Ghost Evacuee" Inflation — INEQUITY

In the absence of mathematical per-capita quota enforcement, vocal or easily accessible camps submit inflated requisitions—stockpiling weeks of food reserves—while isolated rural shelters face starvation. Duplicate requests submitted across shifts drain central warehouses within hours.

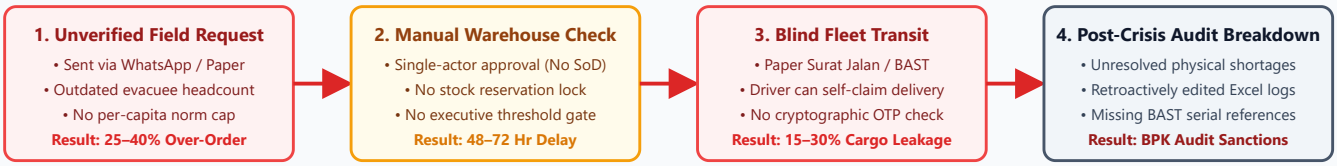
3. Collapsed Segregation of Duties (SoD) — CORRUPTION RISK

Under emergency pressure, the same official often initiates, verifies, and approves high-value aid releases without multi-tier oversight. Paper-based handover receipts (*Berita Acara Serah Terima / BAST*) are routinely signed in bulk, backdated, or falsified by transit personnel without physical verification by camp recipients.

4. Unmonitored SLAs & Modifiable Audit Logs — AUDIT FAILURE

Traditional spreadsheets and standard CRUD databases allow retroactive editing or deletion of transaction records after discrepancies occur. Consequently, Supreme Audit Institution (*BPK RI*) and Inspectorate audits frequently issue formal findings over unaccounted emergency contingency budgets (*Belanja Tidak Terduga / BTT*).

2.3 Structural Failure Cascade (Traditional Paper & Chat Workflow)



2.4 Quantitative Baseline of the Problem

Operational Dimension	Legacy Government Practice (Baseline)	Documented Public Impact	ResQ-Gov Target
Requisition-to-Dispatch Cycle	Paper signatures & WhatsApp escalation across 3 desks	**48–72 hours latency**; misses golden survival window	**≤ 1 hr verify / ≤ 6 hr deliver** (Rule R-07)
Allocation Equity Across Camps	Subjective estimation without per-capita norm limits	**35%+ stock imbalance**; simultaneous hoarding & starvation	**100% Norm-Capped** max 3-day coverage (Rule R-03)
In-Transit Cargo Shrinkage	Paper delivery slips signed without dual-key verification	**15–30% discrepancy** or diversion during surge operations	**0% Unverified Drop-off** via 6-digit OTP (Rule R-05/06)
Post-Disaster Financial Audit	Mutable spreadsheets & unlinked paper BAST folders	Delayed **BPK/Inspektorat** clearance; public distrust	**100% Immutable** Hash-Chain Ledger (Rule R-09)

03 Target User & Market

Multi-Tier Civil Service (ASN) User Personas & Archipelagic Disaster Management Market Sizing

3.1 Primary End-User Personas (Segregation of Duties Architecture)

ResQ-Gov is purpose-built around the statutory organizational structure of Indonesian regional governments (*Pemerintah Provinsi / Kabupaten / Kota*), enforcing **Role-Based Access Control (Rule R-12)** and **Segregation of Duties (Rule R-01)** across four distinct operational personas:

Role & Civil Service Rank	Institutional Unit	Core Operational Responsibilities in ResQ-Gov	Enforced Governance Rules
1. Kepala Dinas ROLE: ADMIN `Dr. Hj. Ratna W.`	**Executive Leadership** Head of Dinas Sosial / Pelaksana BPBD	• High-value requisition authorization (> IDR 25,000,000) • Investigation & resolution of physical delivery discrepancies • Civil apparatus (ASN) account & RBAC management • 1-click cryptographic hash-chain check & official BAST print	`R-01, R-02, R-06, R-07, R-08, R-09, R-11, R-12`
2. Koordinator Logistik ROLE: KOORDINATOR `Ir. Bambang H.`	**Operations Control** Pusdalops PB & Regional Warehouse	• First-tier verification of camp requisitions vs. stock • Inbound aid logging with mandatory BAST/Surat Jalan ref • Fleet & driver dispatch matching cargo weight to truck • Reviewing >30% headcount surge flags & unlocking OTPs	`R-01, R-02, R-04, R-05, R-07, R-08, R-10, R-12`
3. Petugas Posko ROLE: POSKO `Siti Aminah (P01)`	**Field Shelter Unit** Village Evacuation Posts, GOR Shelters	• Mandatory 24h evacuee & vulnerable census update • Submitting norm-capped aid requests (≥30-char reason) • Holding the secret 6-digit Handover OTP for shipments • Performing physical item count reconciliation on arrival	`R-01, R-03, R-05, R-06, R-08, R-10, R-12`
4. Pengemudi Armada ROLE: DRIVER `Budi Santoso (V01)`	**Last-Mile Distribution** Dinas Sosial / BPBD Tagana Fleet	• Receiving shipment manifest & camp coordinates • Executing trip departure (`dalam_perjalanan`) with GPS • Entering Camp Officer's 6-digit OTP at drop-off • Strictly barred from self-confirming final receipt	`R-01, R-05, R-11, R-12`

3.2 Secondary Beneficiaries & External Stakeholders

Displaced Citizens & Vulnerable Groups

Direct humanitarian beneficiaries—especially **infants (*balita*), elderly (*lansia*), disabled persons, and pregnant mothers**—whose nutritional and medical needs are explicitly tracked per camp.

State Auditors (BPK RI & Inspektorat)

Oversight institutions gain instant access to exportable CSV audit trails, cryptographic integrity proofs, and complete BAST chains linking inbound donor batches to final camp receipts.

National Agencies & CSR Donors

BNPB, Kemensos RI, and corporate CSR donors supplying relief batches (e.g., `BAST/BNPB/IX/2026/118`) gain verifiable visibility into how their consignments are allocated.

3.3 Market Sizing (TAM, SAM, SOM) & Deployment Ecosystem

Public Sector Market Funnel (Indonesia & ASEAN)

TAM — Total Addressable Market — **552 Gov Entities + ASEAN**

All **514 Regency/City** + **38 Provincial Governments** in Indonesia (1,104 Dinsos & BPBD agencies), plus ASEAN Member States via AHA Centre.

SAM — Serviceable Available Market — **168 High-Risk Regencies**

168 Kabupaten/Kota classified as "High Disaster Risk" on BNPB's Disaster Risk Index (*IRBI*) across Java, Sumatra, Sulawesi, and Nusa Tenggara.

SOM — Serviceable Obtainable Market (24 Mos) — **15 Priority Regencies**

Initial deployment across **15 flood- and seismic-prone regencies** in West Java, Central Java, and West Sumatra (~600 evacuation camps).

Why Government Demand is Urgent Now — POLICY DRIVERS

- **Presidential Regulation No. 95/2018 on SPBE:** Mandates electronic government administration and auditable digital workflows across all regional agencies.
- **Perka BNPB No. 04/2020 on Disaster Logistics:** Requires standardized inventory accounting, multi-tier authorization, and traceable distribution receipts.
- **BPK RI Audit Scrutiny on BTT Funds:** Regional heads face strict audits over Emergency Contingency Budgets (*Belanja Tidak Terduga*); ResQ-Gov provides turnkey compliance.
- **Zero Hardware Lock-In:** Runs immediately on any smartphone, tablet, or laptop browser already used by ASN officers and drivers.

04 Proposed Solution

End-to-End Disaster Logistics State Machine & The 12 Code-Enforced Governance Rules (R-01 to R-12)

4.1 Architectural Breakthrough: Governance-as-Code

ResQ-Gov transforms disaster relief from an informal administrative workflow into a **deterministic, cryptographically verifiable state machine**. Every mutation must pass through a centralized validation kernel (`src/rules/businessRules.ts` & `src/store/StoreContext.tsx`); any rule violation is **blocked at runtime and recorded as a flagged anomaly in the hash-chained audit ledger**.



4.2 Complete Specification of the 12 Enforced Business Rules (R-01 to R-12)

Rule	Governance Principle	Statutory & Operational Rule Specification	Code-Level Enforcement Mechanism
`R-01`	**Segregation of Duties (SoD)**	Requestor, Verifier, and Approver must be distinct officers. Drivers cannot confirm final receipt.	Runtime actor check (`actor.id !== req.requestedBy`); self-approval blocked & logged.
`R-02`	**Multi-Tier Executive Approval**	Requisitions valued **> IDR 25,000,000** require digital approval from Head of Agency after Coordinator check.	Auto-transitions to `menunggu_persetujuan`; dispatch blocked until `admin` approves.
`R-03`	**Demographic Norm Quota Cap**	Requested qty capped at per-capita norm (max 3-day coverage) minus active in-flight requests for the camp.	Ceiling `maxAllowedQty(item, post) - inFlightQty(...)` enforced on submission.
`R-04`	**Warehouse Stock & Reservation**	Cannot verify or dispatch if unreserved warehouse stock is insufficient. Stock auto-deducts on dispatch.	Checks `availableQty = stock - reservedQty`; creates immutable outbound `MOV` record.
`R-05`	**6-Digit OTP Handover Proof**	Each shipment has a secret 6-digit code known only to destination Posko. Driver must input code on arrival.	**3 failed attempts** lock shipment (`locked = true`) & alert Coordinator.
`R-06`	**Physical Receipt & Discrepancy**	Posko Officer records actual physical qty received per item. If `receivedQty < qty`, status becomes `selisih`.	Raises red executive alert on Kepala Dinas dashboard until Berita Acara resolution.
`R-07`	**Service Level Agreement (SLA)**	**Darurat:** ≤1h verify / ≤6h deliver; **Tinggi:** ≤3h / ≤12h; **Normal:** ≤6h / ≤24h.	Live countdown timer `getSla(req, now)` flags overdue items & historical breaches.
`R-08`	**Mandatory Documentary Evidence**	Rejections require ≥20-char reason; requests require ≥30-char justification; stock-in requires BAST ref.	Form & store mutations reject payloads failing length or BAST reference checks.
`R-09`	**Tamper-Evident Audit Hash Chain**	Every state change & violation is appended to an immutable log hashing its payload + `prevHash`.	No edit/delete API; `verifyAuditChain()` detects any modified or deleted block.
`R-10`	**Evacuee Data Freshness & Surge**	Posko must update evacuees within **24h** before requesting aid. Changes >30% or over capacity are flagged.	Blocks requests if `now - post.lastUpdate > 24h`; sets `flagged: true` on surge logs.
`R-11`	**Session & Account Security**	Auto-logout after **30 mins** idle. **5 failed logins** lock account 15 mins. Min 1 active Admin required.	Enforced by auth guard; prevents disabling the last active Kepala Dinas account.
`R-12`	**Role-Based Access Control (RBAC)**	Each user accesses only routes & data scoped to their assignment (Posko own camp; Driver own vehicle).	Dual-layer enforcement in router (`App.tsx`) and store runner (`run(roles, fn)`).

4.3 Core Mathematical Models Enforced by the Engine

Rule R-03: Net Requestable Quota per Item (i) at Camp (p)

$$Q_{max}(i, p) = \text{ceil}\left(\text{Evacuees}(p) \times \text{Norm}(i) \times D(i) \right) - Q_{inflight}(i, p)$$

where $D(i) = 3$ days if normBasis == 'hari', else 1 (durable)

Rule R-09: Consecutive Cryptographic Audit Block Hash (seq = k)

$$H_k = \text{Hash}(seq_k \mid ts_k \mid actor_k \mid role_k \mid action_k \mid target_k \mid detail_k \mid flag_k \mid H_{k-1})$$

where H_0 = "00000000000000" (Genesis Block Anchor)

05 Key Features & Tech Stack

Six Mission-Critical Functional Modules & High-Performance Zero-Overhead Web Architecture

5.1 Six Core Functional Modules

1. Executive Policy & Multi-Tier Approval Cockpit

ADMIN

Provides the Head of Agency with real-time KPI telemetry (Pending High-Value Approvals, Open Physical Discrepancy Cases, Total Registered Evacuees, and Realized Aid Budget in IDR), regional food/medical stockpile resilience area charts, and a 1-click executive authorization desk for requisitions exceeding IDR 25,000,000.

2. Interactive GIS Crisis & Radar Command Center

SPATIAL GIS

Custom zero-latency SVG geospatial command map visualizing watershed hazard zones (*Zona Bahaya Longsor & Banjir DAS*), evacuation post capacity saturation rings, central Pusdalops warehouse nodes, and live GPS fleet route progress (`0–100%` with dynamic ETA). Includes a live disaster event simulator & dynamic field post registration.

3. Demographic Norm & Quota Requisition Engine

POSKO

Tracks camp headcounts and 4 vulnerable groups (*Balita, Lansia, Disabilitas, Ibu Hamil*). Automatically computes remaining allowable quotas across 9 standardized relief items (Rice 0.4 kg/person/day, Clean Water 2.5 L/day, Instant Noodles 2 packs/day, Medical Kits, Masks, Tents, Blankets, Hygiene Kits, Infant MP-ASI).

4. Warehouse Inventory & Fleet Capacity Dispatcher

KOORDINATOR

Separates physical stock from reserved stock (`availableQty = stock - reservedQty`), triggers low-stock alerts below minimum thresholds, logs inbound donor consignments with mandatory BAST references, and assigns approved shipments to available fleet vehicles (Pick-up 1,000 kg, Truk Engkel 4,000 kg, Truk Fuso 8,000 kg).

5. Zero-Trust 6-Digit OTP Handover & Reconciliation

DRIVER & POSKO

Eliminates transit fraud by requiring the driver to input the destination Posko's secret 6-digit OTP upon arrival. Three failed attempts lock the shipment immediately. Once unlocked/delivered, the Posko Officer performs itemized physical reconciliation, automatically escalating any shortage to `SELISIH` status.

6. Cryptographic Audit Verifier & Official BAST Export

COMPLIANCE

Displays every historical action, actor NIP, timestamp, and `cyrb53` cryptographic hash link. Includes an interactive "*Uji Integritas Rantai*" verifier that validates every block from Genesis (`00000000000000`), plus print-ready official government BAST reports and one-click CSV forensic exports.

5.2 System Architecture & Technology Stack



5.3 Engineering Specifications & Standardized Humanitarian Commodity Catalog

Commodity Code & Name	Category	Unit Price (IDR)	Min Threshold	Sphere / BNPB Per-Capita Norm	3-Day Max Cap (450-Person Camp)
I01 — Beras (Staple Rice)	Pangan	`Rp 14,000 / kg`	`1,000 kg`	`0.40 kg / person / day`	**`540 kg`** per 3-day cycle
I02 — Air Mineral (Clean Water)	Pangan	`Rp 4,000 / L`	`3,000 L`	`2.50 L / person / day`	**`3,375 L`** per 3-day cycle
I03 — Mi Instan & I09 — MP-ASI Balita	Pangan	`Rp 3,500 / Rp 55,000`	`1,500 / 150`	`2.0 packs / 0.10 pkg / day`	**`2,700 packs / 135 pkgs`**
I04 — Paket Obat P3K & I05 — Masker	Medis	`Rp 150,000 / Rp 45,000`	`100 / 80`	`0.05 kit / 0.05 box per person`	**`23 kits / 23 boxes`** (durable)
I06 — Tenda, I07 — Selimut, I08 — Hygiene	Hunian/Sanitasi	`Rp 3.5M / 75k / 85k`	`20 / 500 / 300`	`0.10 tent / 1.0 blanket / 0.25 kit`	**`45 tents / 450 blankets / 113 kits`**

06 Screenshots or Product Demo (Executive & Spatial Command)

Live Product Interfaces Captured Directly from the Working ResQ-Gov Application

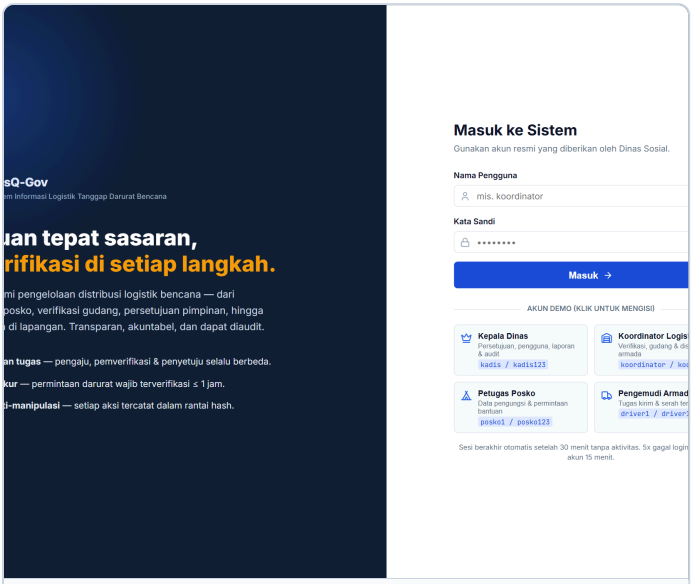


Figure 6.1 — Authentication Portal & 1-Click Role Sandbox (`#/login`)

Enforces session timeout (30 min) and 5-strike brute-force lockouts (Rule R-11), while offering judges 1-click demo credential cards to test all 4 civil service personas (`kadis`, `koordinator`, `posko1`, `driver1`).

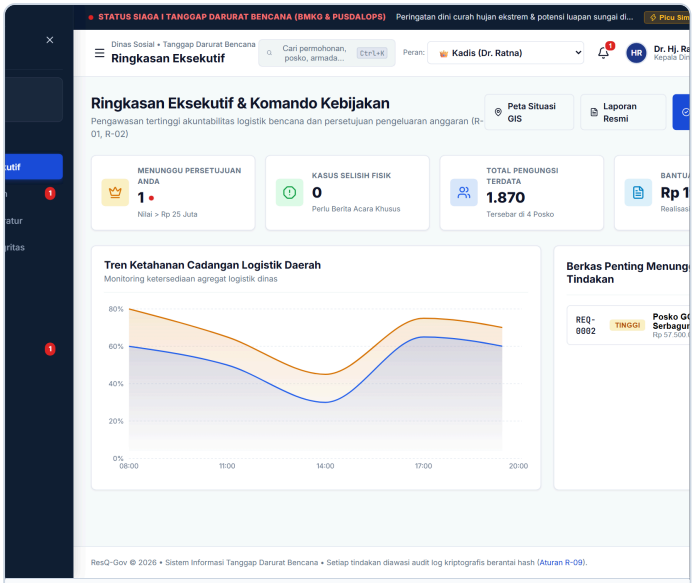


Figure 6.2 — Executive Command Cockpit (`#/admin` — Kepala Dinas)

Real-time visibility into pending high-value approvals (> IDR 25M), physical discrepancy alerts, total evacuees (1,870 across 4 camps), realized aid budget (IDR 19.2M), and regional food/medical stockpile resilience curves.

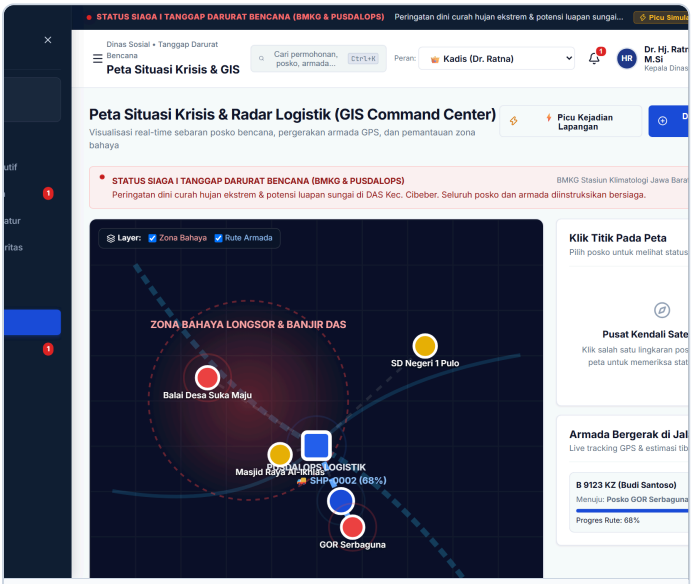


Figure 6.3 — GIS Crisis Situation & Logistics Radar (`#/peta`)

Interactive spatial mapping of watershed hazard zones (*Zona Bahaya Longsor & Banjir DAS*), camp occupancy status rings, Pusdalops warehouse hub, and live GPS truck progress (`SHP-0002` at 68% route completion, ETA 10 mins).

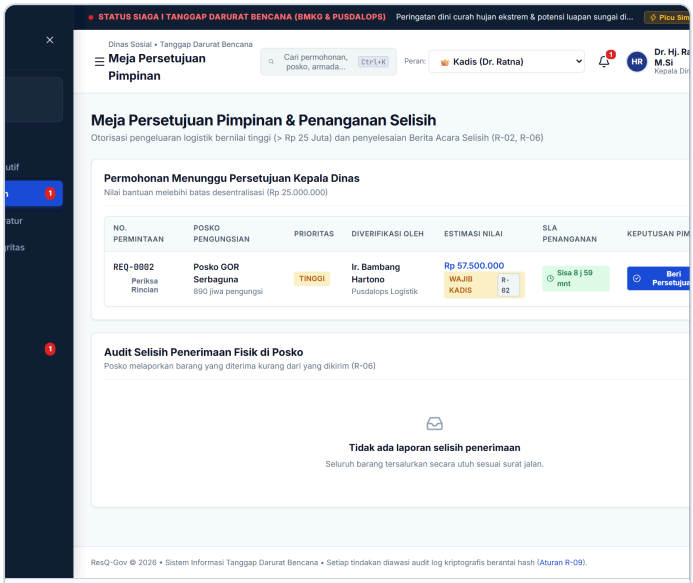


Figure 6.4 — Multi-Tier Executive Approval Desk (`#/admin/persetujuan`)

Enforces Rule R-02: Requisition `REQ-0002` from Posko GOR Serbaguna (10 Refugee Tents + 300 Blankets valued at **IDR 57,500,000**) is automatically held after Coordinator verification until digitally authorized by the Head of Agency.

Top-Bar Live Simulation & Instant Persona Switching for Judges

INTERACTIVE DEMO

Notice the persistent top command bar across Figures 6.2–6.4: (1) **Red BMKG & Pusdalops Early Warning Ticker** broadcasting real-time *SIAGA I* hydrometeorological alerts; (2) **"Picu Simulasi" (Trigger Event) & "Live Sync On" Controls** that inject realistic emergency surges and advance truck GPS positions; and (3) **Instant Role Switcher Dropdown** allowing evaluators to seamlessly experience Segregation of Duties across *Kadis, Koordinator, Posko 1/2, and Driver 1/2* without logging out.

06 Screenshots or Product Demo (Field Verification & Forensic Audit)

Norm-Capped Posko Requisitions, Warehouse Verification, 6-Digit Handover OTP, and Hash-Chain Ledger

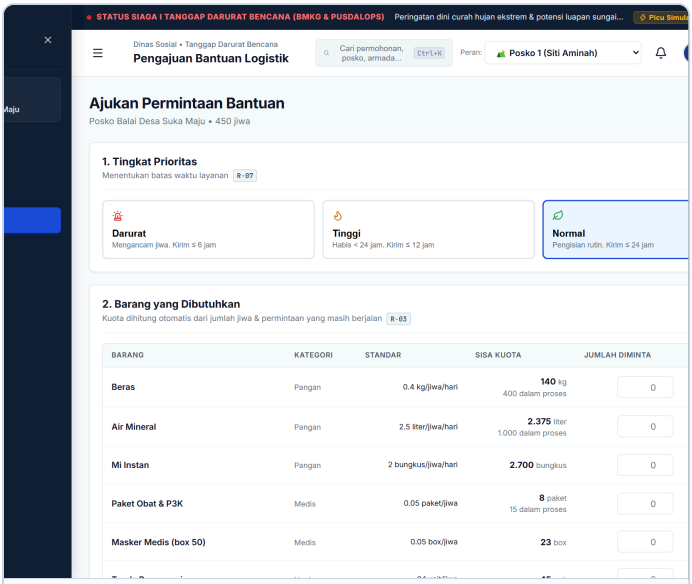


Figure 6.5 — Norm-Capped Requisition Form (`#/posko/ajukan` — Posko)

Enforces Rule R-03 & R-07: Automatically calculates remaining quota (`SISA KUOTA`) for 450 evacuees after subtracting in-flight requests (e.g., 400 kg rice already in process → max 140 kg remaining).

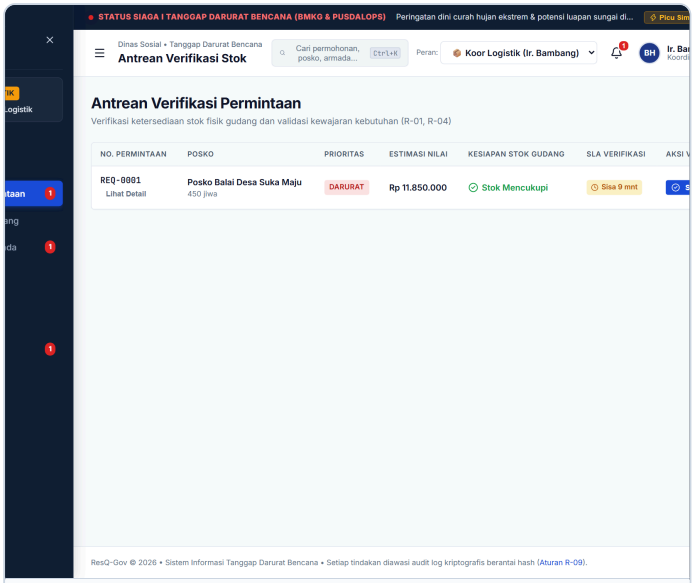


Figure 6.6 — Pusdalops Stock Verification (`#/koordinator/verifikasi`)

Enforces Rule R-01 & R-04: Coordinator Ir. Bambang reviews `REQ-0001` (Darurat priority, IDR 11.85M), verifying real-time warehouse sufficiency (`Stok Mencukupi`) against the 1-hour verification SLA chip.

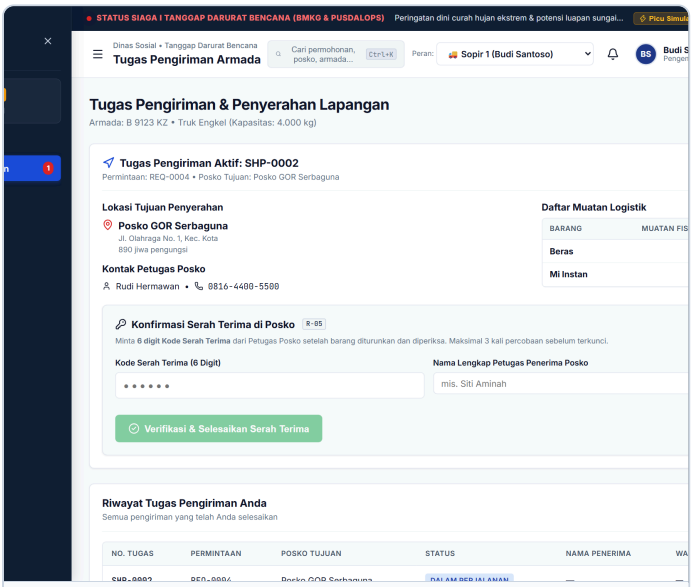


Figure 6.7 — Fleet Driver 6-Digit OTP Handover (`#/driver/tugas`)

Enforces Rule R-05: Driver Budi Santoso (`B 9123 KZ`) delivering 300 kg Rice & 800 packs Instant Noodles to Posko GOR Serbaguna must enter the Posko Officer's secret 6-digit code (`482913`) to complete delivery.

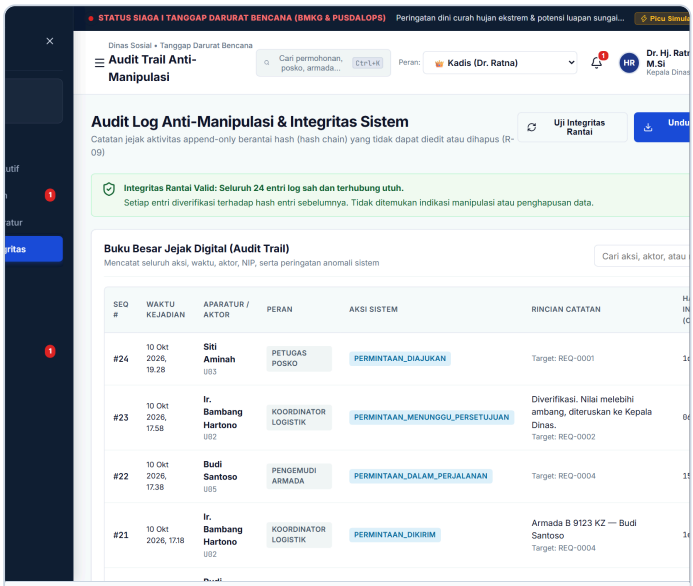


Figure 6.8 — Cryptographic Hash-Chain Audit Trail (`#/admin/audit`)

Enforces Rule R-09: Displays all 24 sequential append-only audit blocks with `cyrb53` integrity hashes and a green banner confirming 100% mathematical chain validity across all transactions.

5-Minute End-to-End Product Demo Script (Video & Live Judging Walkthrough)

DEMO PROTOCOL

Min 0:00–1:15 (Posko)
Login as `posko1`, update evacuee census (R-10), and submit a Darurat request capped by R-03 norms.

Min 1:15–2:30 (Koordinator)
Switch to `koordinator`, verify stock (R-04), escalate >IDR 25M to `kadis` (R-02), and dispatch truck `V01`.

Min 2:30–3:45 (Driver & OTP)
Switch to `driver1`, view GIS route, input 6-digit OTP from Posko (R-05), and reconcile physical count (R-06).

Min 3:45–5:00 (Audit & BAST)
Switch to `kadis`, run cryptographic chain integrity check (R-09), and print official BAST report.

07 Competitive Advantage

How ResQ-Gov Outperforms Legacy Manual Workflows, Commercial ERPs, and Static E-Gov Portals

7.1 Multi-Dimensional Competitive Matrix

Existing disaster logistics approaches fall into three categories: (1) ad-hoc WhatsApp groups and Excel spreadsheets, (2) heavy commercial Enterprise Resource Planning (ERP) suites (e.g., SAP, Oracle, Odoo), or (3) basic government web forms that lack business-rule validation. ResQ-Gov bridges this gap by combining **lightweight, rapid-deployment web engineering** with **strict public-sector statutory enforcement**:

Evaluation Dimension	Manual Paper BAST + WhatsApp / Excel	Commercial Cloud ERP (SAP / Odoo)	Conventional E-Gov Reporting Web	ResQ-Gov Platform (Proposed Innovation)
Segregation of Duties (SoD) Enforcement	None; single officer can write & sign paper slips	Configurable but requires costly custom workflow dev	Role menus only; no actor-level cross-check	**Native R-01 & R-02** (Posko ≠ Koor ≠ Kadis; >IDR 25M gate)
Per-Capita Demographic Quota Capping	None; prone to panic hoarding & duplicate orders	Not built for humanitarian per-capita daily norms	Static number inputs without population math	**Native R-03** (Auto-capped by headcount × norm × 3 days)
Last-Mile Anti-Diversion Handover Proof	Paper signature; easily forged or backdated	Standard barcode scan (driver can self-scan)	Simple "Mark Delivered" button by operator	**Native R-05 Dual-Key 6-Digit OTP** + 3-strike shipment lock
Physical Shortage & Discrepancy Escalation	Buried in paper footnotes; rarely investigated	Inventory adjustment log without executive alert	Not supported	**Native R-06** (Auto-flags `SELISIH` to Head of Agency cockpit)
Audit Trail Tamper Resistance	Very Low (Excel rows easily edited/deleted)	Medium (DBA can alter SQL audit rows)	Low (Standard mutable database table)	**Native R-09 Append-Only Hash Chain** with 1-click verifier
Census Freshness Gate & Surge Detection	Stale headcounts used for days	No evacuee census concept	Manual periodic updates	**Native R-10** (Mandatory ≤24h update + >30% surge flag)
Emergency SLA Monitoring	None; 48–72 hr delays common	Generic ticketing SLA	Timestamps only	**Native R-07** (Live 1h/6h Darurat SLA countdown chips)
Total Cost of Ownership & Setup Speed	Zero license cost, but 15–30% aid loss cost	IDR 500M–2B+ license & 6-month setup	Moderate dev cost; high maintenance	**Zero Per-Seat License**; deploys in <5 mins on any cloud/VPS

7.2 Alignment with APICTA Public Sector & Government Judging Criteria

1. Uniqueness & Technological Innovation

APICTA CRITERION 1

First regional disaster platform to unite **Sphere/BNPB per-capita quota mathematics**, **state-bound 6-digit handover OTPs with brute-force lockouts**, and a **client-verifiable cryptographic hash-chain ledger** inside a zero-dependency-overhead web application.

2. Proof of Concept & Functional Completeness

APICTA CRITERION 2

Not a mockup or slideware: ResQ-Gov is a **100% functional Working Product** featuring 17 interactive views across 4 roles, live cross-tab synchronization, dynamic disaster simulation, print-styled official BAST documents, and CSV audit export.

3. Public Value, Transparency & Accountability

APICTA CRITERION 3

Directly addresses corruption and leakage in emergency public procurement. Every Rupiah of distributed aid is tied to a verified ASN NIP, a BAST reference number, a camp headcount snapshot, and a recipient handover code.

4. Interoperability & National Scalability

APICTA CRITERION 4

Designed for seamless compliance with Indonesia's **SPBE (Perpres No. 95/2018)** and **Satu Data Indonesia** standards, supporting instant deployment on the National Data Center (*PDN*), provincial VPS nodes, or edge servers.

08 Current Status & Development Plan

Technology Readiness Level 7 (Working Product), Operational Dataset & Phased Engineering Plan

8.1 Current Readiness Level: Working Product (TRL 7)

ResQ-Gov has completed full-stack core development and internal stress-testing, achieving **Technology Readiness Level 7 (Working Product / Pilot-Ready System)**. The platform is pre-loaded with a realistic, end-to-end operational dataset modeling a live **SIAGA I Hydrometeorological Emergency in Cibeber & Kota Districts**, enabling comprehensive evaluation of normal, escalated, rejected, and in-transit workflows:

17 Views	1,870	100% Pass	2.79 sec
Complete Role Screens	**Simulated Evacuees**	**Rule & Type Safety**	**Production Build Time**
Admin, Koor, Posko, Driver, GIS	Across 4 active Posko (302 vulnerable)	0 OxLint errors • TS 7.0 verified	228 KB gzipped edge bundle

8.2 Pre-Configured Operational Simulation Dataset (Verifiable in Live Demo)

Entity Domain	Active Records in Seed State	Governance & Edge-Case Scenarios Demonstrated
Civil Service Accounts (`U01–U06`)	6 active personnel across 4 roles with official NIP & rank	Demonstrates RBAC isolation (R-12), SoD cross-checks (R-01), and 1-click persona switching.
Evacuation Camps (`P01–P04`)	Balai Desa Suka Maju (450), SDN 1 Pulo (320), GOR Serbaguna (890), Masjid Al-Ikhlas (210)	Tracks 152 infants (*balita*), 200 elderly (*lansia*), 22 disabled, and 34 pregnant women; enforces 24h census freshness (R-10).
Relief Commodities (`I01–I09`)	9 items across Pangan, Medis, Hunian, Sanitasi	Includes low-stock alert demonstration (`I03 Mi Instan` stock 1,400 < minStock 1,500) and BAST inbound history (`MOV-0001..0007`).
Aid Requisitions (`REQ-0001..0007`)	7 full-lifecycle requests spanning all status states	• `REQ-0001`: Darurat priority awaiting Coordinator verification (SLA countdown) • `REQ-0002`: High-value (**IDR 57.5M**) held at Executive Approval Desk (Rule R-02) • `REQ-0004`: Active transit on Truck `B 9123 KZ` (68% GPS progress, OTP `482913`) • `REQ-0006`: Delivered at camp awaiting physical count reconciliation (Rule R-06) • `REQ-0007`: Rejected duplicate tent request with documented reason (Rule R-08)

8.3 Structured Development & Pilot Implementation Plan

Phase & Timeline	Milestone Focus	Key Technical & Institutional Deliverables	Status
Phase 1 Q2–Q3 2026	**Core Governance Engine & Web Platform**	Domain modeling with BPBD/Dinsos SOPs; implementation of Rules R-01 to R-12; SHA-256/cyrb53 hash-chain audit ledger; 4-role responsive UI & print-ready BAST reports.	100% COMPLETED
Phase 2 Q4 2026 (Current)	**GIS Crisis Radar & Live Simulation Engine**	Interactive spatial map with watershed hazard layers & GPS fleet telemetry; automated emergency event injector; multi-tab real-time state sync; APICTA evaluation readiness.	100% COMPLETED
Phase 3 Q1–Q2 2027	**Backend Persistence, PWA & Regency Pilot**	PostgreSQL + append-only WORM storage backend; Service Worker offline-first sync queue for zero-signal camps; WhatsApp/SMS OTP webhook; Pilot trial in 3 West Java regencies.	SCHEDULED NEXT
Phase 4 Q3–Q4 2027	**SPBE Interoperability & IoT Telemetry**	REST/GraphQL connectors to national SPLP (SPBE), BNPB InaRISK, and Kemensos DTKS; hardware GPS truck transponders & QR/RFID warehouse gate scanners.	PLANNED

09 Business Model

Sustainable B2G & B2B2G Revenue Streams, E-Katalog LKPP Procurement & Fiscal Savings Analysis

9.1 Public-Sector Business Model Canvas (B2G + B2B2G Hybrid)

To ensure rapid institutional adoption without burdening local governments with prohibitive upfront capital expenditure, ResQ-Gov operates a **Dual-Track GovTech Sustainability Model** compliant with Indonesian public procurement regulations (*E-Katalog LKPP*):

Stream 1: Regency B2G SaaS — 60% REVENUE

Annual Subscription & Managed Cloud/PDN Support: Priced at **IDR 75,000,000 / year / Regency (Kabupaten/Kota)** covering unlimited ASN users, evacuation camps, and fleet vehicles. Funded via annual APBD Dinsos/BPBD operational ICT budgets or *Belanja Tidak Terduga (BTT)* preparedness allocations.

Stream 2: Provincial Command — 25% REVENUE

Provincial Multi-Regency Integration License: Priced at **IDR 250,000,000 / year / Province** (or subsidized via BNPB *Dana Siap Pakai* grants). Enables provincial governors and Pusdalops to monitor cross-district buffer stocks, mutual-aid transfers, and regional SLA compliance.

Stream 3: B2B2G CSR Portal — 15% REVENUE

Corporate Philanthropy & ESG Audit Gateway: Priced at **IDR 35,000,000 / year / Enterprise Donor**. Enables FMCG, telecom, and banking CSR donors to track their donated BAST batches down to verified camp handovers for ESG sustainability reporting.

9.2 Value-for-Money & Fiscal Return on Investment (ROI) for a Single Regency

An average disaster-prone regency manages **IDR 5,000,000,000** in annual emergency logistics and social relief supplies across flood and landslide seasons. By eliminating phantom requisitions, duplicate hoarding, and unverified transit shrinkage (conservatively estimated at 15–20% in manual systems), ResQ-Gov pays for its annual subscription more than **10 times over in a single disaster response cycle**:

Fiscal Cost / Savings Component (Per Regency / Year)	Legacy Manual / Paper System	With ResQ-Gov Platform	Net Annual Fiscal Impact
Annual Regional Relief Commodity Budget	IDR 5,000,000,000	IDR 5,000,000,000	Baseline Relief Allocation
Over-Allocation & Hoarding Loss (Est. 12% → 0%)	IDR 600,000,000 wasted	IDR 0 (Capped by Rule R-03)	**+ IDR 600,000,000 Saved**
Transit Shrinkage & Unverified Drop-Offs (Est. 8% → 0.5%)	IDR 400,000,000 lost	IDR 25,000,000 (Caught by R-05/06)	**+ IDR 375,000,000 Saved**
Paper BAST Printing, Courier & Post-Crisis Audit Overtime	IDR 65,000,000	IDR 10,000,000 (Digital BAST/CSV)	**+ IDR 55,000,000 Saved**
Annual Software Licensing, Hosting & ASN Training Cost	IDR 0	IDR 75,000,000 / year	- IDR 75,000,000 Investment
NET ANNUAL PUBLIC VALUE SAVED PER REGENCY	- IDR 1,065,000,000 Leakage	IDR 75,000,000 Subscription	**+ IDR 955,000,000 Net Savings (12.7x ROI)**

9.3 Three-Year Financial & Deployment Projection (2027–2029)

Year 1 (2027) — Pilot Scale — 5 REGENCIES

- **Gov Clients:** 5 Regencies + 1 Province
- **CSR Partners:** 4 Enterprise Donors
- **Gross Revenue: IDR 765,000,000**
- **Public Aid Protected:** ~IDR 30 Billion

Year 2 (2028) — Regional Expansion — 18 REGENCIES

- **Gov Clients:** 18 Regencies + 3 Provinces
- **CSR Partners:** 12 Enterprise Donors
- **Gross Revenue: IDR 2,520,000,000**
- **Public Aid Protected:** ~IDR 105 Billion

Year 3 (2029) — National Standard — 45 REGENCIES

- **Gov Clients:** 45 Regencies + 8 Provinces
- **CSR Partners:** 30 Enterprise Donors
- **Gross Revenue: IDR 6,425,000,000**
- **Public Aid Protected:** ~IDR 265 Billion

10 Impact Plan

Measurable Humanitarian, Fiscal Governance, and Sustainable Development Impact

10.1 Three-Dimensional Public Impact Framework

1. Humanitarian & Citizen Survival

85% Faster

Emergency Verification & Dispatch

Compresses emergency aid verification to **≤ 1 hour** and physical delivery to **≤ 6 hours** (Rule R-07), preventing secondary mortality, dehydration, and untreated injuries in isolated evacuation camps. Explicit tracking of *balita, lansia, disabilitas,* and *ibu hamil* guarantees life-saving nutritional prioritization.

2. Anti-Corruption & Clean Audit

100% Traceable

Cryptographic Chain of Custody

Eliminates "ghost camps" and phantom deliveries through mathematical quota ceilings (R-03), dual-key 6-digit OTP handovers (R-05), and append-only hash chaining (R-09). Directly supports regional governments in retaining **WTP (*Wajar Tanpa Pengecualian*)** audit opinions from BPK RI.

3. Operational & Green GovTech

90% Paperless

Digital BAST & Fleet Efficiency

Replaces multi-carbon paper *Surat Jalan* and manual ledger books with instant digital BAST generation and CSV audit exports. Capacity-matched fleet dispatching prevents overloaded truck breakdowns and reduces unnecessary fuel consumption across hazardous routes.

10.2 Before-and-After Impact Measurement Matrix (KPIs)

Key Performance Indicator (KPI)	Pre-Implementation Baseline	Post-Implementation Target	Verification & Measurement Method
Emergency Verification Latency	12 – 24 Hours (Paper routing)	**≤ 1.0 Hour** (Darurat SLA)	Automated timestamp delta (`verifiedAt - createdAt`) in R-07 engine
End-to-End Aid Delivery Time	48 – 72 Hours	**≤ 6.0 Hours** (Darurat SLA)	OTP completion timestamp (`deliveredAt - createdAt`) in Shipment log
Unaccounted Transit Shrinkage	15% – 30% of total volume	**0% Unrecorded** (100% flagged)	Physical receipt reconciliation (`receivedQty vs qty`) under Rule R-06
Duplicate / Over-Quota Requisitions	25%+ of camp submissions	**0% Permitted** (Hard blocked)	Pre-submission mathematical rejection via `maxAllowedQty()` (Rule R-03)
Audit Trail Preparation Time for BPK	2 – 4 Weeks manual compilation	**< 10 Seconds** (1-Click Export)	Instant CSV ledger export + `verifyAuditChain()` integrity check

10.3 Alignment with United Nations Sustainable Development Goals (SDGs)

SDG 11.5 — Disaster Resilience

GOAL 11

"Significantly reduce the number of deaths and the number of people affected and substantially decrease the direct economic losses relative to global GDP caused by disasters, with a focus on protecting the poor and people in vulnerable situations."

SDG 16.5 & 16.6 — Accountable Gov

GOAL 16

"Substantially reduce corruption and bribery in all their forms" and *"Develop effective, accountable and transparent institutions at all levels"* through code-enforced Segregation of Duties and immutable cryptographic audit logs.

SDG 2.1 & 3.8 — Nutrition & Health

GOALS 2 & 3

Guarantees equitable daily caloric access (0.4 kg rice, 2.5 L clean water, infant MP-ASI) and essential emergency medical/hygiene kits for displaced populations during acute climate shocks.

11 Roadmap & Future Direction

Strategic Evolution from Regency Command Platform to AI-Predictive & ASEAN Interoperable Ecosystem

11.1 Multi-Horizon Strategic & Engineering Roadmap (2026–2028+)



11.2 Deep-Dive into Upcoming Technological Innovations

1. Zero-Internet LoRa & Satellite Mesh Fallback

RESILIENCE

Major earthquakes and flash floods frequently sever cellular base stations (BTS). In Horizon 1–2, ResQ-Gov introduces a **Progressive Web App (PWA) CRDT offline queue** paired with compact LoRaWAN/Starlink Direct-to-Cell gateways, allowing Posko officers and drivers to verify 6-digit HMAC-TOTP handover codes completely offline and sync the hash chain once connectivity returns.

2. AI Predictive Demand & Pre-Positioning Engine

ARTIFICIAL INTELLIGENCE

By ingesting live BMKG rainfall radar, watershed river-gauge telemetry, and historical camp inflow rates, our upcoming **AI Pre-Positioning Module** forecasts 72-hour commodity depletion before stockouts occur—automatically recommending pre-emptive buffer transfers from neighboring district warehouses.

3. BSrE-BSSN Certified Electronic Signatures (TTE)

LEGAL VALIDITY

Integrating **Balai Sertifikasi Elektronik (BSrE - BSSN)** PKI certificates directly into Rule R-02 (Kepala Dinas approval) and Rule R-06 (Official BAST PDF generation), giving every exported ResQ-Gov document full statutory evidentiary force under Indonesia's ITE Law.

4. Cross-Border ASEAN Interoperability (AHA Centre)

REGIONAL SCALE

Standardizing commodity taxonomy and hash-chain manifest schemas to interoperate with the **Disaster Emergency Logistics System for ASEAN (DELSA)** operated by the AHA Centre, enabling seamless cross-border humanitarian relief tracking across Southeast Asia.

National Policy Vision: Towards a Unified Indonesian Disaster Logistics Ledger

Our long-term direction positions ResQ-Gov as the reference open-standard logistics governance module within Indonesia's **National Public Service Portal (INA Digital / SPBE)**—connecting 514 regencies into a single, mathematically governed national resilience network where no displaced citizen is left behind and every Rupiah of humanitarian aid is cryptographically accountable.

12 Team Profile & Contact

Multidisciplinary Core Team, Live System Evaluation Access & Institutional Contact

Lead System Architect

GOVTECH & POLICY

Public Sector Governance & System Architecture Lead

Spearheads statutory translation of BNPB/Kemensos regulations into the 12 binding business rules (R-01 to R-12), Segregation of Duties (SoD) state modeling, and BPK RI audit compliance design.

Lead Software Engineer

FULL-STACK & CRYPTO

Core Platform & Cryptographic Ledger Engineer

Architected the React 19 + TypeScript 7 application, append-only `cyrb53`/SHA-256 hash-chain audit engine, interactive SVG GIS crisis radar, and multi-tab real-time synchronization store.

Operations & UX Analyst

FIELD LOGISTICS

Humanitarian Supply Chain & Field UX Specialist

Designed high-contrast field interfaces for Posko officers and truck drivers under high-stress disaster conditions, Sphere per-capita quota formulas, and the 6-digit OTP handover protocol.

Live Evaluation & Contact Information

- **Innovation Name:** ResQ-Gov (Category 02: Public Sector & Government)
- **Source Repository:** `c:PA2ResQ-Gov` (Vite 8 + React 19 + TS 7)
- **Local / Intranet Demo URL:** `http://localhost:5173`
- **Cloud Readiness:** 100% Static/SPA hash-routed (`#/`) for Vercel, Netlify, or Gov Nginx VPS
- **Contact Email:** `team@resq-gov.id` | `public-innovation@resq-gov.id`

Judge Quick-Login Credentials (1-Click on Login Screen)

DEMO ACCESS

Role Persona	Username	Password	Key Features to Test
Kepala Dinas (Admin)	`kadis`	`kadis123`	Approve >IDR 25M, Hash Audit, BAST Print
Koordinator Logistik	`koordinator`	`koor123`	Verify Stock, Inbound BAST, Dispatch Fleet
Petugas Posko	`posko1`	`posko123`	Update Evacuees, Quota Request, OTP View
Pengemudi Armada	`driver1`	`driver123`	Active Route SHP-0002, 6-Digit OTP Input

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